Xiaoqing PAN

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Re: Postdoctoral research fellow in bacterial cell biology, reference number DR/092/26

Dear Prof. Erhardt,

My name is Xiaoqing, I just finished my PhD study at Hans-Knöll-Institute in Jena, Germany, expect to have my defense in summer 2026. My PhD studied how pathogenic microorganisms produce, process, and deploy RNAs to regulate its biology. Through this work, I developed a strong interest in how pathogens reshape both their own regulatory programs and host responses during stimulations. During my PhD, I investigated RNA regulation in host-fungal interactions, with a focus on fungal small RNAs, dsRNA metabolism, and human innate immune responses. In *Aspergillus fumigatus*, a human fungal pathogen, I analyzed sRNA-seq and mRNA-seq data to characterize the small RNA landscape and the contribution of RNAi machinery to fungal biology. This work gave me practical experience with RNA data analysis, transcriptomics, target-oriented thinking, and the connection between RNA regulation and microbial behavior. In parallel, I analyzed transcriptomic responses of human immune cells after fungal stimulations, further characterizing host alternative splicing and A-to-I RNA editing changing patterns.

I have been reading about the work from your lab on *Salmonella* virulence factors, including flagella, injectisomes, and adhesins, and I find the connection between bacterial motility, secretion systems, and host invasion very compelling. Specifically, the work on sRNA-mediated regulation of flagellin production caught my attention. The connection between trans-acting sRNAs, environmental stress, and virulence-associated T3SS regulation is where I think I could contribute most directly, sRNA IsrJ and its role in epithelial cell invasion (in Dr. Saleh's thesis). The mechanism by which IsrJ promotes invasion remains uncharacterized, and this is a problem I am well positioned to work on. My PhD gave me deep experience in different methods of RNA-seq analysis, small RNA target prediction, RNA secondary structure analysis, and functional validation. Once candidate targets are predicted, I have the tools to test them: luciferase reporter assays, GFP-based localization assays, and GUS reporter assays from my master's work. These are methods I have used to validate RNA-target interactions and gene expression patterns in my own research.

There is a host biology dimension to the IsrJ invasion question that I can also contribute to. I already studied the transcriptomic response of human cells responding to different fungal stimulations, characterizing pathogen-specific signatures in pattern recognition receptor pathways and innate immune signaling. I have hands-on experience with human epithelial cell infections, including immunostaining of A549 cells during *A. fumigatus* confrontation assays. The host side of bacterial invasion, how cells recognize and respond to *Salmonella* T3SS-dependent effector delivery is the aspect I'd like to study beyond fungal invasion. Additionally, my master work on rice-*Xanthomonas oryzae* interactions also gave me early exposure to T3SS biology from the plant bacterial pathology side, where T3SS-delivered effectors suppressing host immunity are a focus.

What draws me to this project more broadly is the opportunity to bring RNA biology tools into a bacterial system where the mechanistic questions are genuinely open. Cutting-edge approaches like advanced live-cell and high-resolution microscopy, RNA-FISH assays are what I want to learn in the future.

Best,

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SUMMARY

Doctoral candidate in Microbiology with deep expertise in small RNA biology, RNAi machinery, and host-pathogen interactions at the RNA level. Experienced in sRNA-seq and mRNA-seq analysis, small RNA target prediction, RNA secondary structure analysis, and functional validation using reporter assays. Hands-on background in microbial genetics, epithelial cell infection models, and innate immune transcriptomics.

TECHNICAL SKILLS

- **Molecular Biology**: sRNA cloning, NGS library preparation, GUS reporter assays, luciferase and GFP reporter assays, Gibson Assembly Cloning, Genetic Engineering, Northern Blot, qPCR, PCR, immunostaining, LC-MS, cultivation of microorganisms (BSL-1/2), plant cultivation and tissue preparation
- **Bioinformatics**: RNA-seq, sRNA-seq, differential expression, alternative splicing, small RNA characterization and target prediction, data visualization, Snakemake, HPC (Slurm), UCSC, SwissProt, KEGG, NCBI
- **Programming**: R, Python, Bash, Git, Docker, Linux
- **Language**: Chinese (native), English (fluent), German (B1)

EXPERIENCES

RNA landscape of human fungal pathogen *Aspergillus fumigatus*

- Characterized the RNAi machinery of a filamentous fungal pathogen: dicer-like protein specificity, argonaute loading, and small RNA biogenesis; performed sRNA cloning, custom sRNA-seq library preparation, target prediction, and RNA secondary structure analysis
- Validated findings using Northern blot, qPCR, and functional reporter assays; generated and characterized knockout and transgenic strains via Gibson assembly
- DOI: 10.1261/rna.079350.122 and DOI: 10.1101/2024.05.24.595671

Human PBMC transcriptome changes upon fungal stimulations

- Analyzed pathogen-specific transcriptomic signatures in human immune cells responding to three fungal pathogens, characterizing PRR pathway activation, alternative splicing, and A-to-I RNA editing
- Built Snakemake-based HPC pipeline; wrote custom R scripts for multi-condition cohort analysis
- DOI: 10.1021/acsinfecdis.4c00598

Human neutrophils exporting extracellular RNA upon *A. fumigatus* confrontation

- Analyzed EV-associated sRNA, tRNA, and mRNA cargo from primary human neutrophils during fungal challenge, including immunostaining of A549 epithelial cells during infection assays

Non-coding RNA analysis of rice-*Xanthomonas oryzae* interaction

- Studied how T3SS-dependent bacterial effectors from *Xanthomonas oryzae* shape host non-coding RNA regulation; performed GUS reporter assays, plant cultivation, tissue preparation, LC-MS, qPCR, and PCR
- DOI: 10.1186/s42483-023-00188-8 and DOI: 10.1016/j.jplph.2022.153905

Teaching

- Microbiology and Molecular Biology, MMB005, 2022-2025

EDUCATION

PhD | Microbiology, Friedrich Schiller University, Germany (supervisor: Dr. Matthew Blango) 2026
Thesis: RNA regulation in host-fungal interactions: investigation of the host epitranscriptome and fungal small RNAs

MSc | Bioengineering, Chinese Academy of Sciences, China (supervisor: Dr. Kuaifei Xia) 2021
Thesis: Non-coding RNAs ciR133 and miR168a-5p play roles in response to Xoo and abiotic stress in rice, respectively

BSc | Hainan University, China 2016

PUBLICATIONS

- **Pan X***, Kelani AA*, Schrettenbrunner L, Prabakar S, Israni B, Blango MG. (2026). An improved description of the small RNA landscape of the human fungal pathogen *Aspergillus fumigatus*. Mol Microbiol 1-12.
- **Pan X***, Bruch A*, Blango MG. (2024). The past, present, and future of RNA modifications in infectious disease research. ACS Infect Dis 10(12), 4017-4029.
- Xia K*, **Pan X***, Chen H, Xu X, Zhang M. (2023). *X00*-responsive transcriptome reveals the role of the circular RNA133 in disease resistance by regulating expression of *OsARAB* in rice. Phytopathol Res 5, 22.
- Xia K*, **Pan X***, Zeng X, Zhang M. (2023). Rice miR168a-5p regulates seed length, nitrogen allocation and salt tolerance by targeting *OsOFP3, OsNPF2.4*, and *OsAGO1a*, respectively. J. Plant Physiol 208, 153905.
- Kelani AA, Bruch A, Rivieccio F, Visser C, Krüger T, Weaver D, **Pan X,** Schaeuble S, Panagiotou G, Kniemeyer O, Bromley MJ, Bowyer P, Barber AE, Brakhage AA, Blango MG. (2023). Disruption of the *A. fumigatus* RNA interference machinery alters the conidial transcriptome. RNA 29(7), 1033-1050.

CONFERENCES

- **2025 - Oral presentation:** German speaking Mycological Society (DMykG), Germany
- **2025 - Poster:** Gordon Research Seminar - RNA Editing, Italy
- **2024 - Oral presentation:** Jena RNA Club Symposium, Germany
- **2024 - Poster:** RNA Society Meeting, UK
- **2023 - Poster:** EMBL - The Non-coding Genome Symposium, Germany

AWARDS & SCHOLARSHIP

- DAAD Scholarship 2021 - 2025
- Jena School for Microbial Communication (JSMC) travel grant 2023

REFEREE

- Dr. Matthew Blango (matthew.blango@leibniz-hki.de)
- Dr. Olaf Kniemeyer (olaf.kniemeyer@leibniz-hki.de)